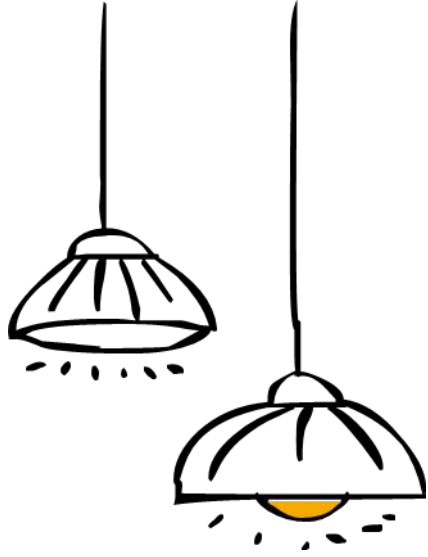
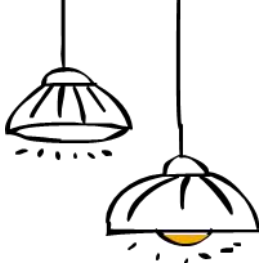




Data Types in Business Analytics

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Lecture Outline

- Explore the importance of data type
- Compare different types of business data
- Explain the data analysis tips
- Discuss business data case studies



Why Data Types Matter?

- **Data's DNA:** Before analyzing data, a manager must understand its "DNA."
- **Proper Analysis:** Data types dictate which statistical tests are valid and which visualizations are appropriate.
- **Garbage In, Garbage Out:** Incorrectly identifying a data type leads to flawed models.
- **Storage & Cost:** Different types require different storage architectures (SQL vs. NoSQL).



Quantitative (Numerical) Data

- Quantitative data represents measurable quantities.
- It is further divided based on the "fineness" of the measurement.
- **A. Discrete Data**
- **B. Continuous Data**



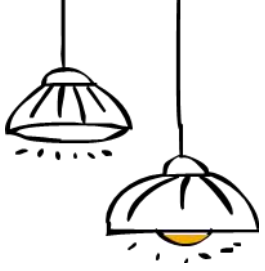
Discrete Data: the countable data

- Definition: Countable values with no "in-between" (integers).
- Discrete data represents values you can count one by one.
- No decimals or fractions are allowed.
- Usually used for counting people, objects, or events.
- Example: the number of students in a class (30, 31, 32) or how many books you have (5, 6, 7)
- Business Example: Number of customer complaints, daily store visitors, or inventory units.



Discrete Data – Analytic Tips

- Visualized with Bar Charts because values are separate and countable.
- Frequency tables work well for summarizing how frequent each value occurs.
- Often analyzed using counts, proportions, or percentages.
- Good for comparing categories or distinct groups.
- Example analyses: number of customers per day, number of items sold in each category.



Continuous Data: Measurement

- Continuous data represents values you can measure.
- These values can take any value within a range, including decimals.
- Can be infinitely precise depending on the measuring tool.
- Usually used for measuring time, amount, distance, or size.
- Example: someone's height (150.5 cm), or the temperature (29.8°C)
- Business Examples: sales revenue amount, delivery time in hours, product weight, machine operating time, etc.

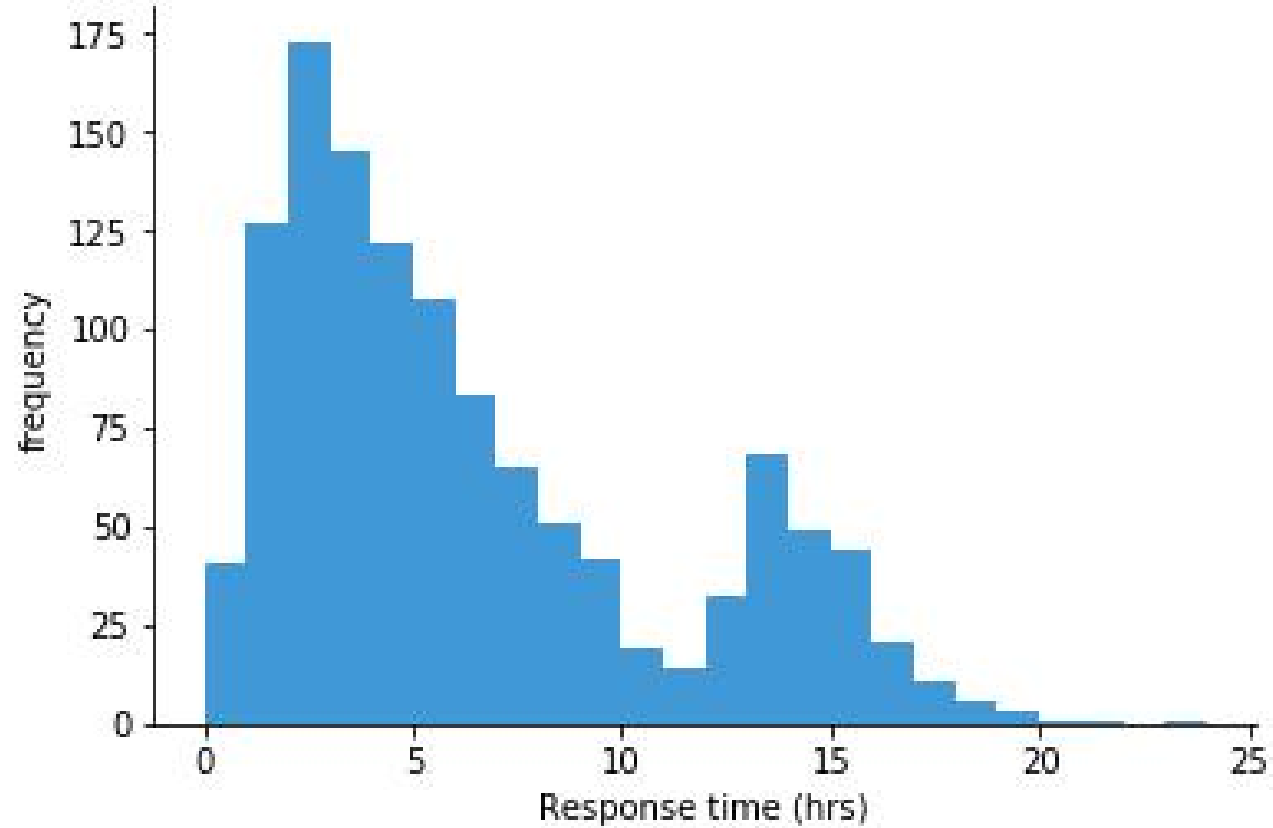


Continuous Data – Analytic Tips

- Visualized with Histograms or Line Charts - the data flows across a range.
- Descriptive statistics (mean, median, standard deviation) are applied.
- Useful for understanding trends, patterns, and distributions.
- Often used in forecasting and modeling - the values change smoothly.
- Example analyses: sales revenue over time, temperature trends, processing time in minutes.



Histogram for Continious data





Qualitative (Categorical) Data

- Qualitative data refers to information that describes qualities, characteristics, or categories rather than numbers.
- They help us understand what type or which group something belongs to.
- They are not measured with numbers; instead, they represent labels or categories used for classification in business.
- **A. Nominal Data**
- **B. Ordinal Data**



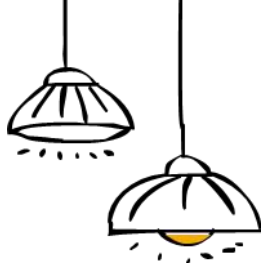
Nominal Data (Labels Without Order)

- Nominal data represents names or labels with no ranking or order.
- Categories are different, but none is “higher” or “lower.”
- Common in business for identifying customer groups, product categories, or market segments.
- Examples: customer gender, product brand, payment method, industry type.



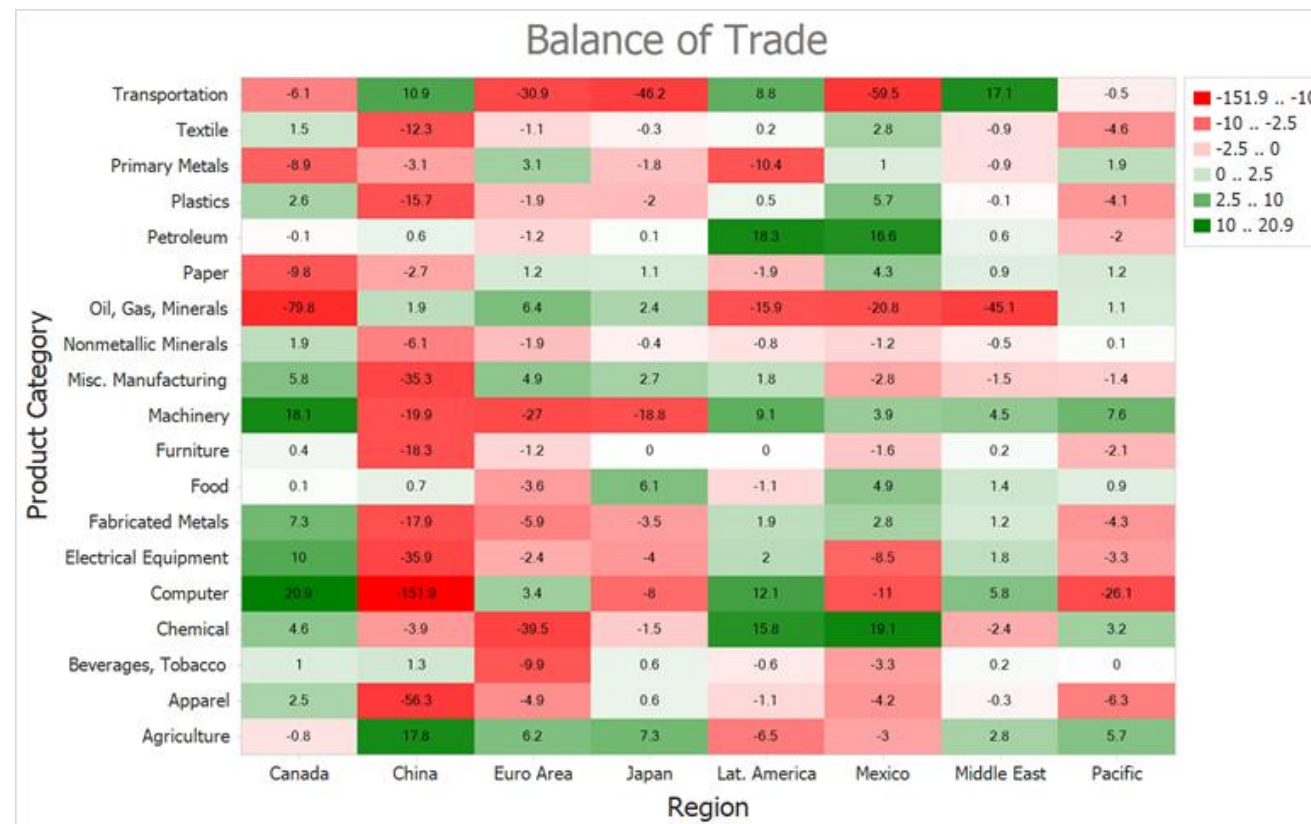
Nominal Data - Analytic Tips

- Usually visualized with Bar Charts or Pie Charts or Heatmaps because each category is separate.
- Best summarized using frequencies and percentages.
- Works well for customer segmentation, brand comparison, and preference analysis.



Heatmaps in Nominal data

- Example: Usage: Cross-tabulating two categorical variables (e.g., Region vs. Product Line)



Source: raw.githubusercontent.com



Ordinal Data: Ordered Category Data

- Ordinal data uses categories, but these categories have a meaningful order.
- The order matters (e.g., “High” is above “Medium”), but the gaps between levels are not equal.
- Common in business for ratings, satisfactions, rankings, and priority levels.
- Examples: satisfaction levels (Poor → Fair → Good → Excellent), education level, service priority.

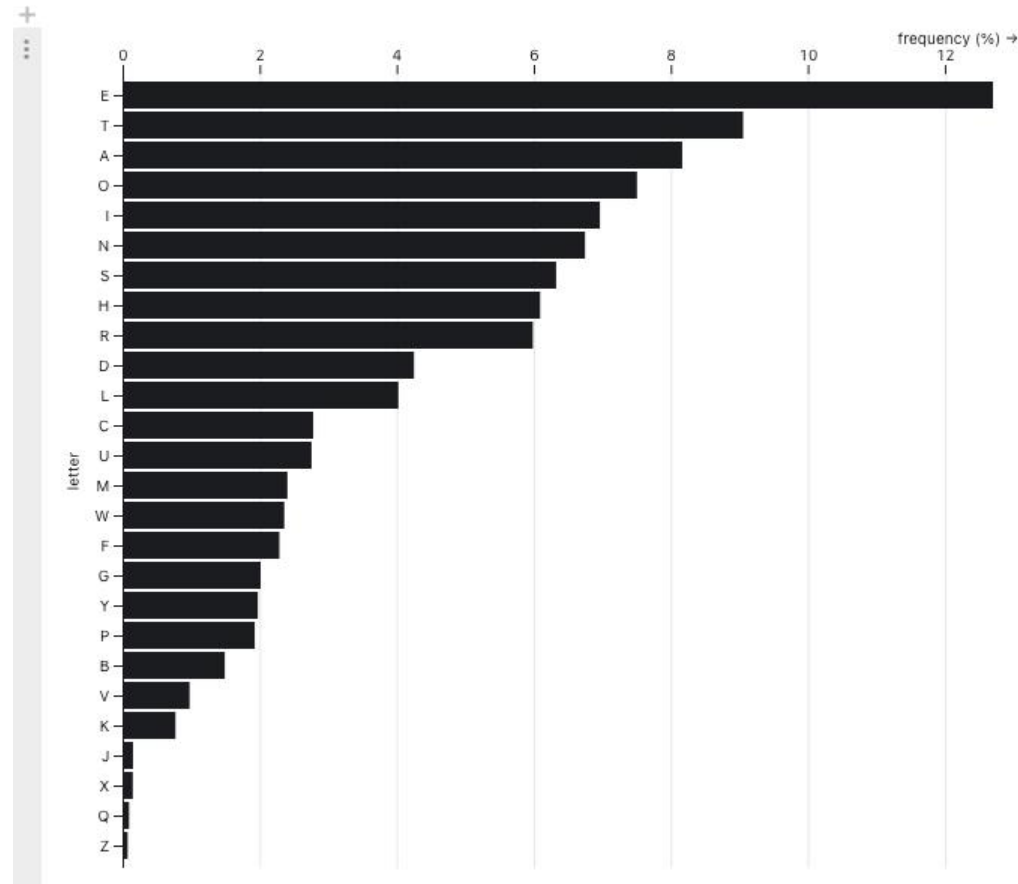


Ordinal Data - Analytic Tips

- Usually visualized with Ordered Bar Charts to show ranking clearly.
- Best described using the median or mode, not the mean.
- Useful for analyzing customer satisfaction, service quality, and preference intensity.



Ordered Bar Charts for Ordinal data



Source: observablehq.com



Advanced Classification: The 4 Levels of Measurement

- To perform advanced analytics, we use Stevens' Levels of Measurement.
- Classify data into four types based on how much mathematical meaning each type carries.

A. Nominal

C. Interval

B. Ordinal

D. Ratio

- As we move from Nominal → Ordinal → Interval → Ratio, the data become more powerful for statistical analysis.



Nominal Level (Category Labels Only)

- Represents names, labels, or groups with no order.
- Only shows difference, not ranking.
- Best summarized using counts and percentages.
- No arithmetic possible.



Ordinal Level (Ordered Categories)

- Represents categories that have a meaningful order, but differences between levels are not measurable.
- Analyze with median or mode.
- Cannot calculate mean meaningfully.



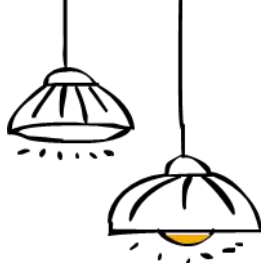
Interval Level (Equal Intervals, No True Zero)

- Numeric data where the difference between values is meaningful, but there is no true zero point.
- Zero does not mean “nothing.”
- Examples: temperature in Celsius or Fahrenheit, dates, calendar years.
- Use Line Charts and Histograms.
- Can calculate mean and standard deviation.

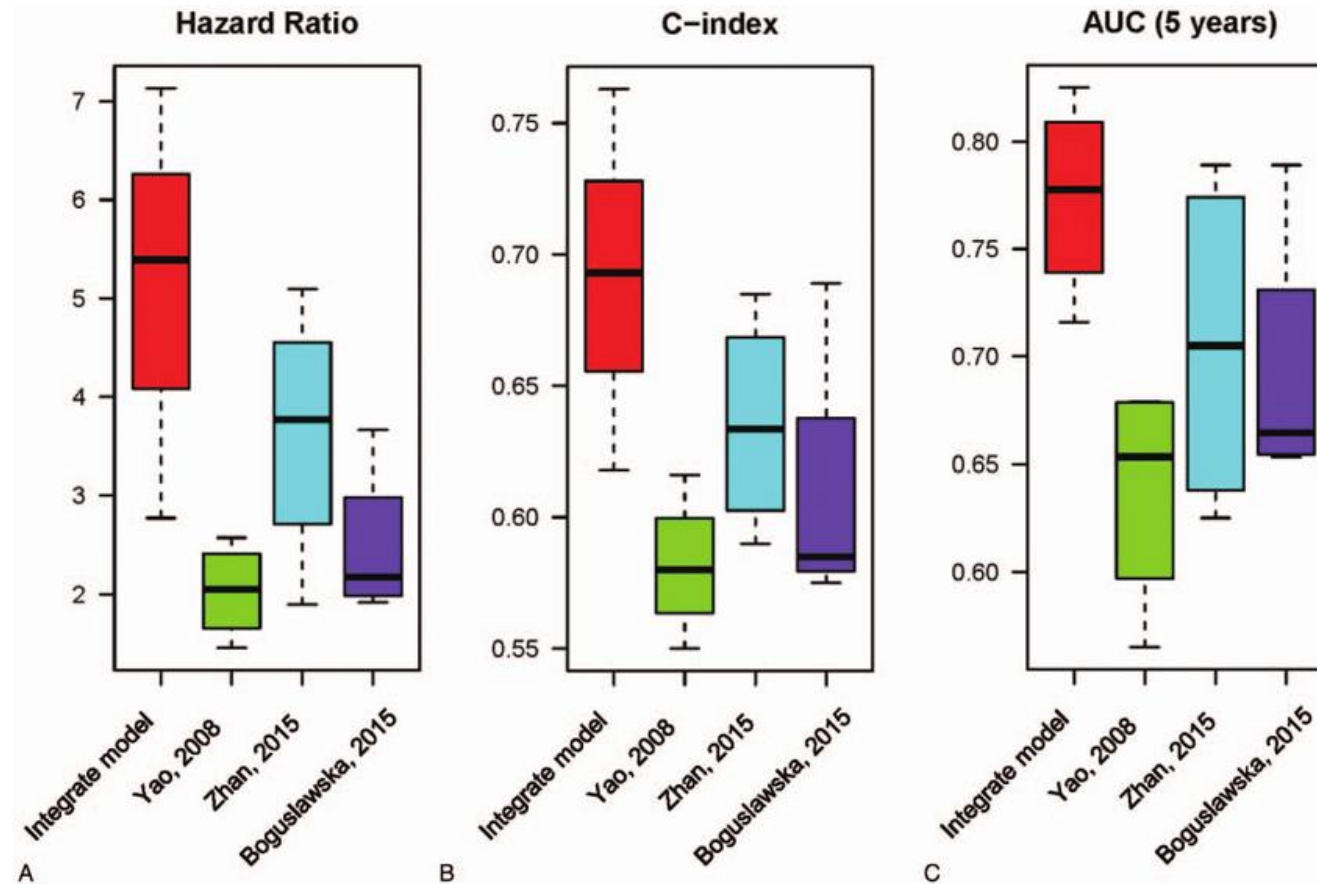


Ratio Level (Highest Level – True Zero Exists)

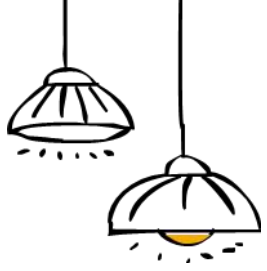
- Numeric data with equal intervals and a true zero, meaning zero represents “none.”
- Ratios are meaningful (20 kg is truly twice 10 kg).
- Examples: revenue, weight, height, age, distance, number of sales.
- Visualized with Histograms, Line Charts, Box Plots.
- Supports all statistical techniques: mean, variance, correlations, regression.



Box plot for Ratio data

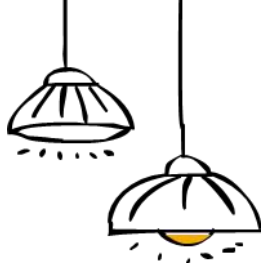


Source: https://www.researchgate.net/figure/Box-plot-of-3-indicators-hazard-ratio-HR-C-index-AUC-in-different-datasets-A-Box_fig2_328675442



Comparison

Level	Characteristics	Operations	Business Example
Nominal	Labels only	Counting, Mode	Product Categories
Ordinal	Order matters	Median, Percentiles	Star Ratings (1-5)
Interval	Order + Equal gaps	Addition/ Subtraction	Temperature, Dates
Ratio	Order + Gaps + True Zero	Multiplication/ Division	Revenue, Salary, ROI



The Visualization Selection Matrix

Target Analytics	Data Type	Recommended Chart
Comparison	Nominal / Ordinal	Bar Chart, Clustered Bar
Composition	Nominal (Parts of a whole)	Stacked Bar, Donut Chart
Distribution	Continuous	Histogram, Box Plot
Relationship	Ratio / Interval	Scatter Plot, Bubble Chart
Trend	Time + Ratio	Line Chart



Unusual Data Structure

- In business and data analytics, data comes in different forms and formats, and understanding these types is crucial for making effective decisions.
- Broadly, data can be quantitative or qualitative, and for practical analysis, it is often classified as unusual structures.
- **Structured, Semi-structured, and Unstructured**
- This classification helps us determine how the data can be stored, accessed, and analyzed.



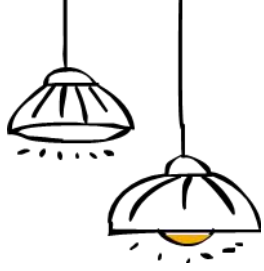
Structured Data

- Structured data is information that is organized, searchable, and stored in fixed formats such as tables, rows, and columns.
- It fits neatly into databases and spreadsheets, making it easy for analysts to query and run calculations.
- In business, this includes data like sales numbers, customer IDs, transaction records, inventory lists, and performance metrics.

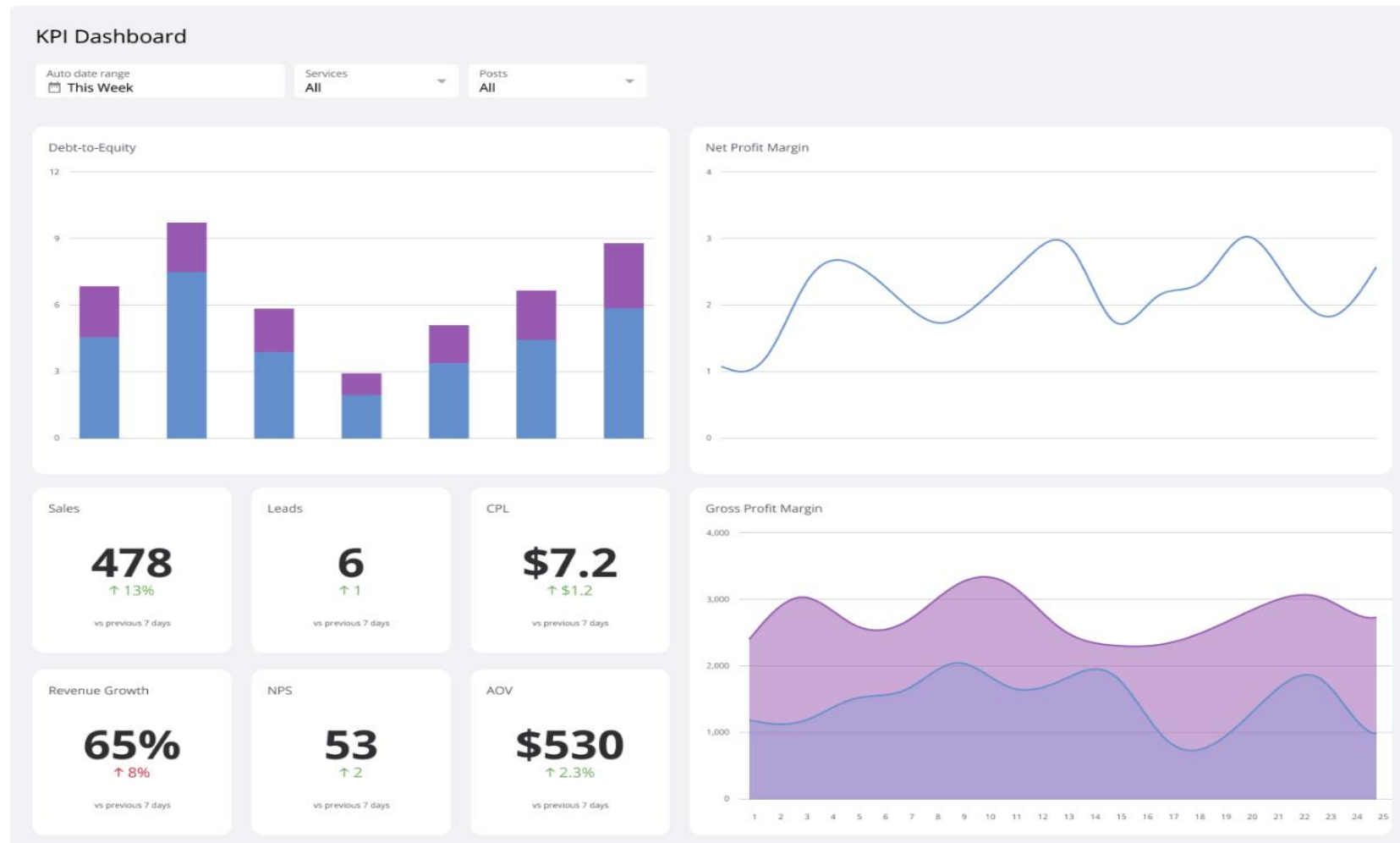


Structured Data - Analytic Tips

- Best analyzed using SQL queries, dashboards, and statistical models.
- Works well with BI tools because it is clean, consistent, and machine-readable.
- Easy to visualize using tables, bar charts, line charts, and KPIs dashboard.



KPI dashboard



Source: [images.klipfolio.com](https://www.klipfolio.com)



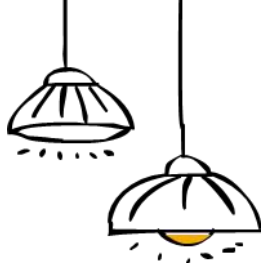
Unstructured Data

- Unstructured data has no fixed format, making it harder to store, search, and analyze directly.
- This includes text, images, audio, video, emails, customer reviews, chat messages, and social media posts.
- Businesses use unstructured data to understand customer behavior, sentiment, and emerging patterns that are not captured in numeric tables.

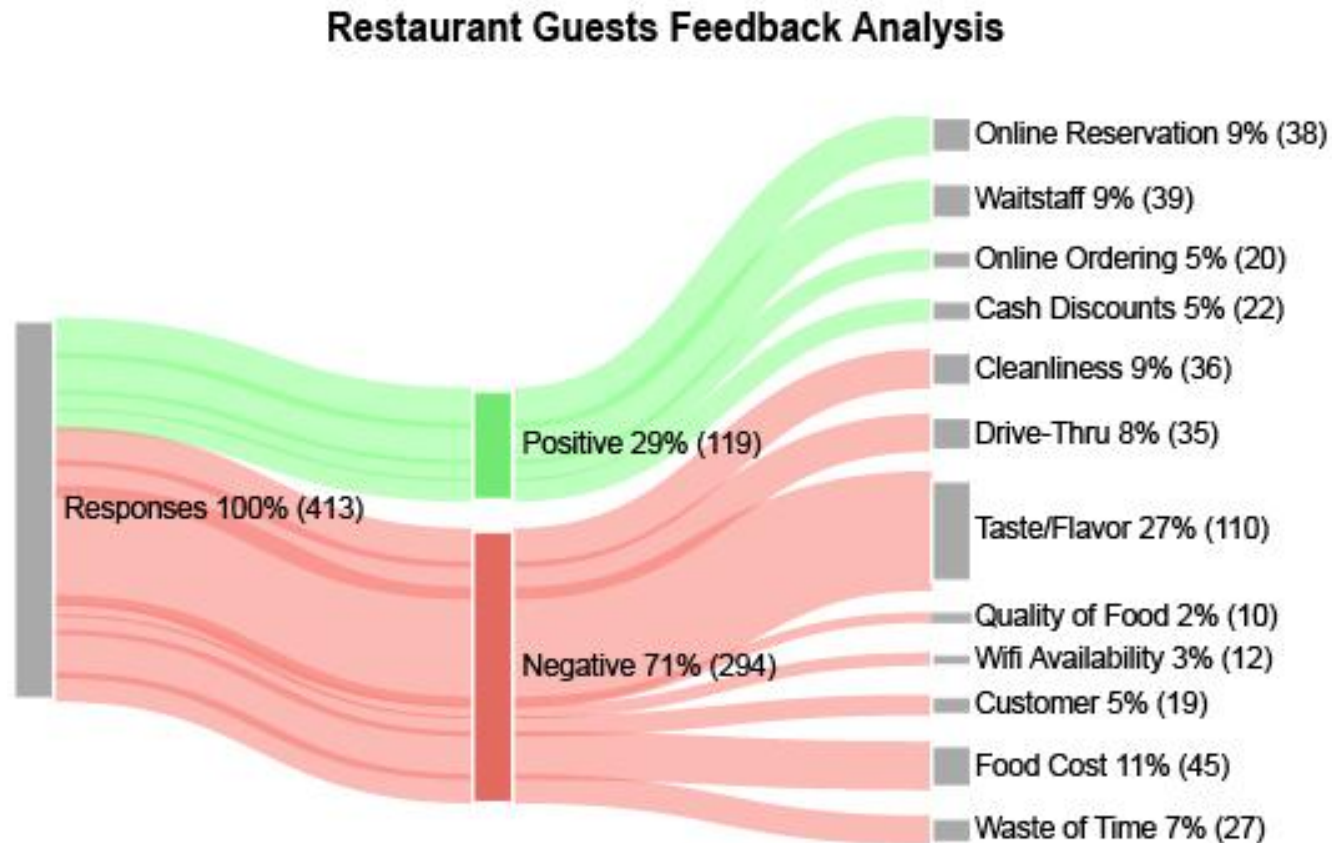


Unstructured Data - Analytic Tips

- Requires advanced analytics such as natural language processing (NLP), text mining, image recognition, or AI-based tools.
- Useful for discovering customer emotions, trends, opinions, and hidden insights.
- Best visualized through word clouds, sentiment charts, tag clusters, or dashboards combining text and numbers.



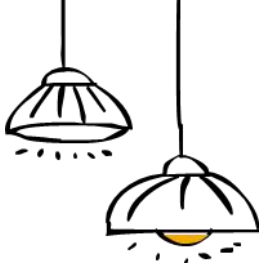
Sentiment Charts in Unstructure data





Semi-Structured Data

- Partially organized; has some structure but not fixed like a database table.
- Examples: XML files, JSON data, emails with headers, online forms.
- Analytic Tip: Can be parsed and analyzed with Python programming or BI tools.
- Visualize using tables, charts, or dashboards after extraction.



Comparison of Unusual data

Feature	Structured Data	Semi-Structured Data	Unstructured Data
Format	Fixed rows & columns	Partially organized	No fixed format
Storage	Databases, spreadsheets	XML, JSON, emails	Text, images, audio, video
Ease of Analysis	Easy	Moderate	Hard
Examples	Sales, inventory, IDs	Email headers, XML, JSON	Social media posts, videos
Visualization	Tables, charts, dashboards	Tables, charts, dashboards	Word clouds, sentiment charts, AI dashboards
Tools for Analytics	SQL, BI tools, Excel	Parsing scripts, BI tools	NLP, AI, Machine Learning



Discussion Questions

- The Question: An HR Director wants to track "Level of Education" (High School, Bachelors, Masters, PhD) and "Current Salary." What is the most accurate classification for these?



Discussion Questions

- Question: An HR Director wants to track "Level of Education" (High School, Bachelors, Masters, PhD) and "Current Salary." What is the most accurate classification for these?
- **Answer:**
- Education has a logical rank (Ordinal), and
- Salary is a measurable amount with a true zero (Ratio).



Discussion Questions

- Context: You are a consultant for a global retailer. You are looking at a spreadsheet containing "Shipping Method" (Express, Standard, Economy) and "Shipping Cost" (\$).
- The Question: If the CEO asks for a report on which shipping method is the most cost-efficient? Please types of data are there?



Discussion Questions

- Context: You are a consultant for a global retailer. You are looking at a spreadsheet containing "Shipping Method" (Express, Standard, Economy) and "Shipping Cost" (\$).
- Question: If the CEO asks for a report on which shipping method is the most cost-efficient, how do you handle these two data types?
- **Answer:**
- Shipping Method is Ordinal (there is a clear hierarchy of speed/priority).
- Shipping Cost is Continuous.



Discussion Questions

- Context: Your HR department wants to predict which employees are likely to quit. They give you a dataset with: Employee Tenure (Years), Department, and Last Performance Rating (1–5).
- The **Question**: Which of these variables can you perform math on (like calculating an average), and which can you only use for grouping?



Discussion Questions

- **Context:** Your HR department wants to predict which employees are likely to quit. They give you a dataset with: Department, and Last Performance Rating (1–5).
- **Answer:**
- Department is Nominal; you can only use it to group/filter (e.g., "Is turnover higher in Sales?").
- Performance Rating is Ordinal.



Discussion Questions

- Question: Your HR database assigns every employee a unique ID number (e.g., #102, #103). If you calculate the average of these ID numbers, is the result meaningful? Why or why not?



Discussion Questions

- Question: Your HR database assigns every employee a unique ID number (e.g., #102, #103). If you calculate the average of these ID numbers, is the result meaningful? Why or why not?
- Answer: No. Even though they are digits, Employee IDs are Nominal. They are labels, not measurements. Averaging them would be like averaging social security numbers; it provides no insight into the "average employee."



Discussion Questions

- Question: A marketing team collects customer reviews and comments from social media about a new product. What type of data is this?



Discussion Questions

- Question: A marketing team collects customer reviews and comments from social media about a new product. What type of data is this?
- **Answer:**
- This is Qualitative - Unstructured Data because the information is text-based, descriptive, and does not have a fixed format.



THANK YOU

Please email me if you have any queries